# Jun Hau Chang


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### Mechatronics Engineering Student ‣ Focus: Engineering Internship

Energized by hands-on exploration and real-world problem-solving

- Curious, driven, and hands-on individual with a passion for problem-solving and a direct, practical approach to challenges. Thrives in **dynamic environments** and **adapts quickly** to new tasks and learning styles.
- Demonstrated strengths in multitasking, critical thinking, and professional conduct through academic and internship roles. Known for **fast learning** and creative thinking under pressure.
- Experienced working within real-world engineering environments through a **biomedical software internship**, contributing to software **maintenance**, structured **debugging**, **verification**, and **system architecture** development.
- **Multilingual communicator** (English, Mandarin, Malay) with strong written and verbal fluency. Experience in **peer tutoring** and **note-taking support roles**, showcasing interpersonal skills and adaptability.
- **Permanent Resident** (subclass 132) with full working rights in Australia. **Available** for a **full-time** job in summer.

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## Education and Training

**Master of Engineering (Honours):**
Smart Manufacturing — Expected in 11/2027
Monash University | Clayton, VIC

**Bachelor of Engineering (Honours): (Currently Year 4| WAM:80)**
Robotics and Mechatronics, Major in Artificial Intelligence — Expected in 06/2027
Monash University | Clayton, VIC

**Monash University Foundation Year (WAM:93)** — 07/2023
Sunway College | Selangor, Malaysia

**The International General Certificate of Secondary Education (IGCSE)** — 05/2022
UCSI International School (High School) | Selangor, Malaysia

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## Technical Skills

### Programming

- Python
- C, C++, C#
- Java
- MATLAB
- MIPS (MARS)

### Software & Application Development

- .NET Framework
- Windows Presentation Foundation (WPF)
- Model–View–ViewModel (MVVM) architecture
- Graphical User Interface (GUI) development
- Debugging and software maintenance

### Software Tools

- GitHub
- Jira
- Linux (Ubuntu)
- ROS
- OPC UA

**Embedded & Hardware**
- Arduino IDE
- STM32 (CubeIDE)
- HDL (FPGA)
- PeguinPi

**Simulation, Design & Documentation**
- SolidWorks (CAD modelling)
- LTspice
- MS Office Suite

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## Experience

### Monash Connected Autonomous Vehicles (MCAV) — Member
2/2026 - present

Monash University | Clayton VIC 3168

- Contributing to the development of an autonomous vehicle from the ground up, with a focus on software, simulation, control, and automation systems.
- Developing and testing autonomous vehicle functions using ROS, RViz, Gazebo, and Linux-based tools for simulation, visualisation, and system integration.
- Working on key autonomous driving capabilities including path planning, SLAM, lane detection, object avoidance, and vehicle control within a team-based engineering environment.

### Internship: Software Vacation Student
11/2025 – 2/2026

Invetech | Mount Waverley VIC 3149

- Contributed to software maintenance and feature development within an established biomedical motion-system codebase.
- Resolved compiler and static-analysis warnings, fixed software bugs, and implemented GUI controls following existing design patterns.
- Applied structured debugging techniques to trace root causes and implement reliable, non-regressive fixes.
- Performed verification checks to ensure changes met Jira task specifications and did not impact existing validated functionality.
- Developed scripts and supported software testing for electromechanical test rigs, contributing to test automation activities.
- Gained architectural understanding of OPC UA–based client–server systems through documentation review, UML modelling, and analysis of reference implementations.
- Worked within industry-standard workflows including Jira task tracking, code reviews, and quality documentation in a regulated biomedical environment.

### Casual Work: Note Taker for Student with Disabilities
2/2025 - present

Monash University | Clayton VIC 3168

- Took detailed and accurate notes during lectures to support students with disabilities.
- Ensured notes were well-organized, easy to follow, and delivered on time.
- Collaborated with staff to meet students' individual learning needs.

### Volunteering Work for Kaamatan Festival
7/2025

Kingston City Hall | Moorabbin, Australia

- Assisted in preparing ingredients, including cutting vegetables and portioning food.
- Helped with dishwashing and maintaining a clean food preparation area.
- Served food and drinks to guests in a friendly and respectful manner.
- Supported cultural celebrations by working with a diverse volunteer team.

### Math Tutor
11/2023 - 02/2024

Eye Level | Selangor, Malaysia

- Supported classroom activities by helping kids understand difficult concepts in mathematics.
- Adapted teaching style according to individual learning styles of each kid.
- Provided individualized tutoring sessions to help kids enhance their math skills.
- Instructed kids on proper techniques for solving problems in Mathematics.

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## Academic Project

### Industrial SCADA Monitoring System

- **Researched SCADA** applications for industrial automation and process monitoring.
- **Designed** a conceptual SCADA architecture integrating **PLCs**, **sensors**, and **HMI** for real-time monitoring.
- Evaluated **system components**, **communication** methods, and **operational monitoring** considerations for industrial environments.
- Produced a **system design proposal** demonstrating industrial automation principles.

### Embedded Sensor Design and Signal Processing (TRC3500)

- Designed and built multiple **sensing systems** from the ground up, including **soil moisture**, **pin-drop impact**, and **respiratory rate sensors**.
- Developed analogue **signal conditioning** circuits incorporating amplification, filtering, and noise reduction to improve measurement accuracy.
- Interfaced sensors with embedded hardware for real-time digital signal acquisition.
- Applied **digital signal processing** techniques to extract, analyse, and interpret sensor data.
- Implemented **data logging** and visualisation for performance evaluation and experimental validation.
- Performed calibration, testing, and error analysis to improve sensor accuracy, repeatability, and robustness.

### PenguinPi Autonomous Fruit Collection Robot

- Developed an autonomous robot capable of detecting, classifying, and collecting fruit using **YOLO-based object detection** and **EKF-based SLAM**.
- **Developed and trained** a custom **YOLOv5 model** with data augmentation and tuning to achieve robust fruit recognition performance across lighting and occlusion scenarios.
- Implemented **A*** path-planning algorithms for navigation and dynamic obstacle avoidance.
- Designed a **PID-based control system** for smooth waypoint tracking and fruit approach.
- Integrated camera vision, and robot kinematics into a real-time perception–planning–control pipeline.

### Warman competition

- Collaborated in a multidisciplinary team of 5 (mechanical, electrical, and software) to design and build a **task-performing robot**.
- Led integration of **mechanical systems** and **control logic**; handled both physical assembly and system programming.
- **Coordinated testing sessions** to ensure subsystem compatibility and functionality.
- Robot successfully completed competition tasks during trials; enhanced cross-functional teamwork and technical communication skills.

### Audio Synthesis Circuit Design

- Designed and built a **complete audio signal** processing pipeline, from microphone input to speaker output.
- Collaborated on **preamp circuit** using electret mic and op-amp (TL974) to condition signals for STM32 ADC input.
- Integrated **STM32 microcontroller** running DSP binary to perform audio filtering and signal conversion.
- Connected STM32 DAC output to LM386 amplifier and headphones, **fine-tuned audio quality** through iterative testing.

### True RMS-to-DC Converter Design

- Designed and built an analogue circuit to compute the **true RMS value** of AC and DC signals.
- Developed sub-circuits for rectification, square-law conversion, low-pass filtering, and logarithmic square-root computation.
- Conducted **LTspice simulations**, breadboard implementation.

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## Accomplishments and Honours

- MathWorks Course Completion Certificate
- SOLIDWORKS CAD Design Associate Certificate
- Monash University Foundation Year: Peer Mentor in Peer Mentoring Programme
- Monash University Foundation Year: Monash High Achiever award
- HSK Proficiency Test: Achieved level 5
- UCSI International School: Valedictorian Award
- UCSI International School: Student council member (Treasurer)